Robustness Enhancement in Image Classification via Data Augmentation

Authored by: Jagriti Koirala, Pritesh Das

The University of Texas at Arlington

Abstract*—In this paper we study the effect of different augmentation techniques on clean accuracy as well as robustness of the image classifiers on CIFAR-10, CIFAR-100 and SVHN datasets. The ResNet-18 classifiers are used, and robustness is estimated through the CIFAR-10-C (corrupted version of CIFAR 10) dataset. As a baseline augmentation method, the pipeline with random cropping and horizontal flipping of images was compared with the application of Mixup and CutMix methods as well as augmentation with synthetic images generated using GANs in case of CIFAR-10. During the middle phase of experiments, which included shorter training periods, strong settings of Mixup and training of the models for 10 epochs, the use of any augmentation technique considerably diminished clean accuracy and had no effect on robustness for a few corruptions from CIFAR-10-C. In the final phase, the training period increased up to 20 epochs for CIFAR-10 and CIFAR-100 and to 15 epochs for SVHN while the learning rate scheduling became the cosine one. A Milder Mixup setting was applied because of which the difference between the performance of the baseline and Mixup almost disappeared. CutMix demonstrated comparable performance regarding clean accuracy but lower corruption accuracy on CIFAR-10-C. Using augmentation of CIFAR-10 with images synthesized by GAN and labeled using pseudo-labeling led to improved corruption accuracy without degrading clean accuracy.

I. INTRODUCTION

Although modern classifiers tend to attain near-perfect accuracy rates on benchmarks, their robustness tends to suffer considerably in the face of real-world distortions such as noise, blur, compression artifacts, and others. In other words, modern image classifiers lack generalizability – a feature that allows the model to achieve the same level of performance despite some input variability. As such, the current research seeks to explore both clean and corrupted accuracy of the ResNet-18 model, trained on three distinct datasets (CIFAR-10, CIFAR-100, and SVHN) and evaluated in terms of CIFAR-10-C robustness, with the latter still keeping the computational budget manageable even in terms of a commodity GPU (such as Google Colab).

While data augmentation is commonly used as a means to achieve better generalizability, its influence on the model robustness may differ greatly depending on the particular algorithm used. As the name implies, augmentation changes the training distribution but keeps the labels intact in the effort to learn a decision boundary that ensures higher accuracy rates in a wider variety of situations. For the purposes of this research, a baseline consisting of random cropping and horizontal flipping will serve as an example of a classical data augmentation method based on label-preserving augmentation, while two additional methods – Mixup and CutMix – will be used to compare the former against label-mixing approaches capable of interpolating images/labels and producing smooth decision surfaces. Finally, data generated by a generative adversarial network (GAN), specifically a DCGAN on CIFAR-10, will be used to determine whether pseudo-labeled synthetic data might be helpful for increasing robustness without sacrificing accuracy rates on clean images.

One of the main research questions revolves around the ability of these three data augmentations to provide improvements in the tradeoff between accuracy and robustness, especially in the context of datasets of different levels of complexity. Another important question relates to the ability of the generated GAN images – despite being imperfect – to improve robustness without decreasing clean accuracy. Finally, the need for this research stems from the fact that the augmentation itself does not behave uniformly: preliminary experiments performed in this course showed that using a very short training regime, along with moderate Mixup parameters resulted in lower accuracy on all datasets compared to the baseline, as well as no increase in robustness against some corruptions on CIFAR-10-C dataset.

In terms of relevant literature, this project builds upon prior research into data augmentation-based robustness enhancement, augmentation–corruption interaction, GAN training using limited amounts of data, differentiable augmentations, as well as comprehensive surveys of mix methods such as Mixup and CutMix. Classical spatial transforms (random cropping and horizontal flipping) form the basis of the regularization technique employed in this paper. Mixup and CutMix are aggressive label-mixing techniques intended to produce smooth decision boundaries, which have been used in prior studies as a means to improve generalizability of a neural net. Instead of reproducing any single methodology exactly, this paper employs a controlled student-project approach, involving the use of one backbone architecture (ResNet-18), an optimizer family and a limited number of datasets that help isolate effects of each augmentation configuration.

The corresponding notebook proceeds according to the aforementioned staged research plan. First, the intermediate phase establishes a CIFAR-10 baseline augmented by standard augmentation and incorporates Mixup to that end. Then Mixup pipeline is replicated for CIFAR-100 and SVHN, limited CIFAR-10-C corruption testing is conducted, as well as DCGAN skeleton on CIFAR-10 is built. Building upon these results, the final phase introduces explicit reproducibility controls, cosine learning rate decay, Mixup α adjustment, longer training time for all datasets involved, CIFAR-10-C corruption evaluation of 19 corruption types and five severities. Finally CutMix and GAN augmentation are applied to the classification problem on CIFAR-10 and evaluated accordingly.

II. LITERATURE REVIEW

There has been a plethora of research on using data augmentation techniques as an effective tool to increase both the accuracy and robustness of deep image classifiers. For example, Rebuffi et al. find that well-chosen data augmentation policies can help achieve robustness against common corruptions at no significant loss of accuracy, which suggests that it is important to select appropriate augmentations in light of expected test-time corruptions. In another study, Mintun et al. explore the interaction between certain augmentations and various corruption types to discover that choosing proper augmentation types can sometimes have non-trivial detrimental impacts on robustness if not appropriately calibrated to the distribution of test-time corruptions. Thus, these two lines of work make it necessary to compare augmentations on benchmark datasets where corruption types are predefined.

There has also been growing interest in using generative adversarial networks (GANs) to synthesize additional data samples in order to enhance classifier training in data-scarce settings. Specifically, Zhao et al. examine the impact of using image augmentations during GAN training on generating diverse samples for improved classifier performance. Meanwhile, Karras et al. and Hong et al. introduce two different ways of GAN training on limited data by applying augmentations throughout both generator and discriminator pipelines to make training more stochastic. Additionally, Agustsson et al. suggest that it is possible to further enhance the efficiency of GAN training with the aid of mask GAN architectures in case of very limited labeled samples. Yang et al. summarize previous research on GAN-based data augmentation and empirically confirm that while synthetic data generated by a trained GAN may benefit classifier performance, gains heavily depend on sample consistency and fidelity. Thus, GAN augmentation is an interesting alternative to classical pixel-space augmentations worth exploring.

Finally, there have been numerous surveys of deep CNNs as state-of-the-art architectures for image classification. All these surveys consider ResNet architectures among others as a powerful and robust baseline to build upon when exploring image augmentations and robustness. Moreover, Zhang et al. conduct a survey on interpolation-based data augmentation techniques, including the widely used Mixup technique and its modifications, and classify all existing augmentation strategies based on their approach to mixing inputs. It has been shown that these augmentation techniques can improve robustness on top of improving classifier accuracy in computer vision problems.

III. DATASETS AND EXPERIMENTS SETUP

Three datasets are used in this paper: CIFAR-10, CIFAR-100, and SVHN. CIFAR-10 contains 60,000 32×32 color images divided equally between 10 classes (airplane, car, bird, cat, deer, dog, frog, horse, ship, truck), with 50,000 training images and 10,000 test images (6,000 per class). CIFAR-100 features the same resolution but 100 classes organized into 20 "superclasses", with 600 images (500 train + 100 test) per class, providing a more difficult challenge in comparison to CIFAR-10 due to higher classification complexity and lower sample number. The SVHN (Street View House Numbers) dataset consists of RGB photographs of street house numbers (digits 0-9) obtained from Google Street View, with 73,257 training images and 26,032 test images; there are also 531,131 auxiliary training samples for SVHN. While the SVHN images have the same dimensions as CIFAR images, the former are significantly more difficult due to varying backgrounds and lighting conditions. Collectively, those three datasets make it possible to analyze how the results of augmentations apply to different difficulties and visual characteristics of the task.

PyTorch data loaders are created for all datasets with training and test subsets strictly separated. CIFAR-10 and CIFAR-100 provide 50,000 training images and 10,000 test images each, while SVHN adds 73,257 training images and 26,032 test images, based on the setup used throughout the rest of the paper. To estimate robustness against corruptions, CIFAR-10-C, an analog of CIFAR-10, in which 19 corruption types of different levels (in total 5) are applied to test images is used. ResNet-18 architecture is selected as the basis of the classifier, with the last layer being resized according to the respective number of classes which was 10 in the case of CIFAR-10 and SVHN, 100 for CIFAR-100). At the midway point, the models are fine-tuned using PyTorch and torchvision with Adam optimizer and cross-entropy loss for the relatively short training procedure. For the final implementation, the same backbone network architecture is used, although the training procedure was enhanced by using fixed seeds for reproducibility, cosine-annealing learning rates schedule, and tracking history for further analysis.

Default data augmentation scheme is composed of random crops with padding and random horizontal flips during the training, while the only preprocessing step performed at inference time is tensor conversion. Mixup augmentation technique involves creation of convex combinations of two training samples along with linear label mix, where a mixing ratio λ from the β distribution is applied, where α is chosen according to empirical tuning on CIFAR-10 and then reused for CIFAR-100 and SVHN; when cutmix is used, the patch of one image is replaced by another image and labels mixed accordingly with a corresponding patching probability.

DCGAN type architecture is selected for GAN training. It uses fully connected layers and transposed convolutions to map a 100 dimensional latent noise vector to a 32x32x3 output tensor, employing Tanh as an activation function for this process. The discriminator architecture consists of consecutive strided convolutions and a sigmoid prediction layer implemented using a logistic loss function. During the midway phase, the two networks are trained simultaneously for 50 epochs using Adam optimizer and BCEWithLogitsLoss (Binary Cross Entropy Loss). A grid of four generated images shows that a blurry image of a CIFAR-10 like image is successfully produced. For the final part of this project, these synthetic images are pseudo-labeled with the help of the CIFAR-10 converged baseline classifier and used to create an augmented dataset.

IV. MODELS AND METHODOLGIES

The experiment followed a progressive setup where initially a baseline ResNet-18 model was trained using typical augmentations in order to obtain benchmark results. Additionally, Mixup was integrated with the identical architecture as well as training process in terms of its main parameters, so that any discrepancies between the two models would be associated mainly with augmentation techniques. Evaluation of robustness properties involved the use of CIFAR-10-C, where initially only a small sample from this dataset at the severity level 3 was evaluated, while later a comprehensive study covered all corruption levels of all classes in this dataset.

As for the baseline model, its training procedure adhered to the traditional supervised learning pattern, namely a simple iteration over the dataset, cross-entropy loss, and mini-batch updating, in conjunction with Adam optimization. On the other hand, Mixup training included generation of augmented samples by combining two inputs in the mini-batch and then computing weighted loss using the sampled lambda coefficient, which makes Mixup directly comparable to the baseline since their architectures, training data sets, and data loader structures remain unchanged.

Three primary improvements were made for obtaining more accurate results during the final stage of the project. Specifically, the first improvement refers to using deterministic CuDNN and fixed random seeds for ensuring repeatability of experiments, while the second involves collecting statistics regarding such metrics as loss, testing accuracy, and learning rates per epoch, which facilitates obtaining clear plots and tables. Finally, the third improvement was replacing short-run settings and poor learning schedules used at the previous stage with cosine learning rate annealing, as initial experiments involving Mixup were underperforming compared to the baseline.

Additionally, two branches of augmentation methods have been explored. Specifically, the first method involves CutMix, which consists of selecting one random rectangular patch of one image and replacing it with the same-sized patch from another image, followed by weighting losses by the respective patches' areas. The second augmentation technique refers to using GANs in order to generate pseudo-labelled CIFAR-10 synthetic images using a baseline classifier and concatenating those with the actual CIFAR-10 data set to train a classifier again. As for the implemented GAN, it is unconditional rather than being class conditional, hence pseudo labeling was necessary.

V. RESULTS

A. Midway Results

The midway experiments intended to set a functional baseline on CIFAR-10 and later apply the same pipeline to CIFAR-100, SVHN and evaluate corruption robustness on CIFAR-10-C. On CIFAR-10, a ResNet-18 trained for five epochs with conventional data augmentation (random cropping with padding and horizontal flips) showed test accuracy of 69.8%. Addition of Mixup augmentation with $\alpha = 1.0$ resulted in test accuracy of 55.9%, suggesting that in this setting Mixup drastically hindered model performance relative to the baseline. The same architecture and transformations were applied to CIFAR-100 with the final fully connected layer adjusted to 100 classes. Three epochs of training produced the baseline test accuracy of approximately 24.0%, whereas the test accuracy after introducing Mixup with $\alpha = 1.0$ dropped to 21.5%. Consequently, just like on CIFAR-10, Mixup in this setting showed no improvement for shorter training periods. SVHN comprised 32×32 RGB images. In its turn, the baseline ResNet-18 model trained for three epochs reached test accuracy of approximately 80.60%, while adding Mixup with $\alpha = 1.0$ decreased SVHN accuracy to 67.38%.

For assessing robustness properties, two CIFAR-10 models were evaluated on CIFAR-10-C corrupted with three different distortions (Gaussian noise, shot noise, defocus blur) with severity level 3. The baseline model achieved approximately 64–67% test accuracy across corruptions, whereas the Mixup model showed consistently lower scores of 52–58%, implying no robustness benefits from Mixup in midway settings. Finally a lightweight DCGAN was used as a generative model in experiments on CIFAR-10, producing blurry although class-like 32 × 32 images. This suggests that in this case the generator has learned a rough representation of the data distribution, although it had not been used to augment the training of the classifiers yet.

B. Final Results

The final phase included experiments in three different directions: (i) longer training schedules for all datasets, (ii) hyperparameter tuning and robustness evaluation on CIFAR-10-C for Mixup and (iii) additional augmentations (CutMix and synthetic data generation).

1) Clean Accuracy on CIFAR-10, CIFAR-100 and SVHN

In order to examine the effect of longer training, the number of epochs was increased to 20 for CIFAR-10 and CIFAR-100 and 15 for SVHN in combination with cosine annealing learning rates. With that, the baseline ResNet-18 achieved noticeably higher accuracies on all datasets. On CIFAR-10, the baseline model achieved 82.94% test accuracy, whereas the best performing Mixup model (tuned around $\alpha = 0.2$) reached 82.41%. Therefore, unlike in the midway experiments, the gap between the models almost disappeared after longer training and moderate intensity of Mixup augmentation.

Similarly, the baseline model on CIFAR-100 produced 53.83% test accuracy, while the corresponding Mixup model scored 53.34%, thus almost not differing from the baseline. Thus, in this configuration and setting, Mixup does not lead either to improved or deteriorated clean accuracy on CIFAR-100 relative to the baseline.

For SVHN, the best-performing models were the baseline and the one with Mixup ($\alpha = 0.2$). The baseline achieved 93.82%, whereas Mixup resulted in 93.71% test accuracy, i.e., similar behavior as on CIFAR-10 and CIFAR-100. From training curves obtained from a consolidated history file, one can see that all models kept on improving with further training, and Mixup models usually showed slightly higher training loss, yet achieved the same accuracy in tests as non-Mixup models.

2) Robustness on CIFAR-10-C

For assessing robustness, each model was tested on the CIFAR-10-C benchmark with the calculation of both per-corruption accuracies and mean corruption accuracy (mCA) for 19 corruption types and five severity levels. In this respect, for the baseline and tuned Mixup ($\alpha = 0.2$) models, the mCA on CIFAR-10 amounted to 72.57% and 72.51%, respectively, implying near-perfect similarity in the models' robustness profiles. When applying CutMix with $\alpha = 1.0$ as an augmentation technique to the CIFAR-10 classifier, the mCA value decreased to 67.14%, meaning a significant robustness loss for this setting in spite of similar clean accuracies.

However, the picture becomes clearer by looking into per-corruption accuracies. In particular, for most corruption types (e.g., Gaussian noise, shot noise, snow, frost, and fog) the averaged accuracies for the Mixup model (with $\alpha = 0.2$) stayed within ±1% from the baseline one and sometimes were even better (contrast and glass blur). Thus, as expected, the robustness profile of the Mixup model is almost the same as that of the baseline one with regard to many types of distortions. CutMix, in its turn, underperforms relatively often across noise and blur distortions.

One may attempt to increase robustness by augmenting training data with GAN-generated samples pseudo-labeled by a converged baseline model. The result of this approach on CIFAR-10 reached a slightly higher mCA value of 72.85%, thus outperforming both the baseline and the Mixup models on CIFAR-10-C without reducing clean accuracy on the original test dataset. The analysis of per-corruption accuracy values shows slight, yet consistent increases in blurs and noises for the "baseline + GAN synthetic" classifier, which may indicate a positive influence of augmentation data diversity via GAN generation and pseudolabels.

3) *Summary of Midway vs. Final Results*

Summarizing results from the final phase, one should note a noticeable difference between the midway and final results. At midway, Mixup with α = 1.0 and a few epochs of training substantially decreased clean accuracy on CIFAR-10, CIFAR-100 and SVHN and failed to produce any robustness gains when evaluated on CIFAR-10-C. Tuning parameters to α = 0.2, applying cosine annealing, and training Mixup models for more epochs allowed achieving test accuracies on par with the baselines on CIFAR-10, CIFAR-100, and SVHN. These results suggest that the performance of Mixup models is very dependent on the training regime and hyperparameters. CutMix with α = 1.0 demonstrated competitive clean accuracy, but notably lower mean corruption accuracy. Finally, adding synthetic data generated by GAN and pseudo-labeled by the baseline model helped boost the mCA value slightly above both the baseline and Mixup models.

VI. DISCUSSIONS

Results suggest that the effects of augmentation on both the test accuracy of classifiers and their robustness depend heavily on the training protocol and the specific values of augmentation hyperparameters. In particular, at the midstage, all Mixup variants involved α = 1.0 with no more than 3–5 epochs, which resulted in inferior accuracy of all classifiers on CIFAR-10, CIFAR-100, and SVHN compared to the baselines and failed to make the models more robust to any of the corruptions included in CIFAR-10-C dataset. Evidently, this negative effect is caused by overly aggressive interpolation in combination with insufficient training time needed to disentangle highly interpolated samples and learn the true class structures.

As the number of epochs increased in the final stage, combined with a decrease of α to 0.2, Mixup showed quite a different pattern. With training for 20 epochs on CIFAR-10 and CIFAR-100 and for 15 epochs on SVHN, Mixup approached baseline test accuracies almost perfectly, thus closing previously observed large gaps. Moreover, on CIFAR-10-C, the average corruption accuracy of the Mixup (α = 0.2) model was statistically indistinguishable from the baseline (72.51% vs. 72.57%). The analysis of individual per-corruption scores showed small oscillations around baseline values for various types of noise, blur, weather, and digital corruptions. Therefore, we may conclude that, in the case of a particular architecture trained on our datasets, Mixup acts as a relatively neutral regularization method in a properly set regime: it neither significantly boosts robustness nor incurs large sacrifices of accuracy.

CutMix produced a different balance between the two characteristics. In contrast to Mixup, CutMix training with α = 1.0 on CIFAR-10 provided similar clean accuracy as baseline training but led to much worse corruption accuracy (67.14% vs. 72.57%). The analysis of per-corruption curves showed that, in particular, the decline affected several classes of noise and blur corruptions. This result indicates that, due to the spatially localized nature of CutMix, there may be a preference for learning discriminative patches instead of global features, hence resulting in better in-distribution performance and poor out-of-distribution generalization. Indeed, it is well-known from literature that certain patch-based augmentations can increase in-domain accuracy while making a classifier sensitive to a change of input distributions.

Another way to achieve positive results through data augmentation is to train classifiers using generated samples labeled via some oracle model. For the case of DCGAN used as a generator on CIFAR-10, it turned out that augmenting the dataset with such images slightly improved mean corruption accuracy (72.85%) without decreasing the clean accuracy. In particular, the per-corruption curves demonstrate modest improvements of some blur and noise transformations. It should be noted that, despite being very low quality, blurry and distorted, DCGAN-generated images provide additional diversity to the distribution of training samples, helping the classifier to generalize better. Thus, augmentation with pseudo-labeled GAN-generated images can be viewed as a way to make the classifier less prone to specific corruption distributions in the testing environment.

An important lesson derived from the experiments above is that more augmentation does not necessarily mean more robustness. By comparing the mid-stage and final-stage performances of the same augmentation approach, we found that a significant change in training protocol can convert a negative effect into a neutral one and vice versa. Moreover, the comparison of different augmentations (Mixup, CutMix, GAN-based augmentation) shows that they have quite different robustness profiles when clean accuracies are similar. For practical applications the message to take away from this paper is that augmentation for robustness purposes should be considered as an optimization problem in itself.

Finally we want to emphasize that, when evaluating models, it is crucial to use corruption benchmarks alongside conventional test sets. As seen from the presented results, on CIFAR-10, all final-stage models were indistinguishable by clean accuracies, whereas CIFAR-10-C showed a significant variability among the CutMix, Mixup, baseline, and GAN-augmented training protocols. In general, we believe that evaluation of models in potentially corrupted environments should be a mandatory part of the evaluation pipeline when deploying classifiers under non-ideal conditions.

VII. CONCLUSION

This paper investigates the effects of different types of data augmentation techniques on the accuracy and robustness of the ResNet-18 classifier trained on three benchmark datasets (CIFAR-10, CIFAR-100, and SVHN) as well as the corruption benchmark (CIFAR-10-C). Experimental findings show that the effect of applying augmentation techniques heavily depends on the set of hyperparameters used and the training regime. In particular, at an early stage, combining Mixup with a relatively large coefficient and limited training leads to a considerable performance drop. Meanwhile, at a later stage, after training for an extended period of time, and using a smaller mixing coefficient, Mixup approaches the same level of both clean and corruption accuracies as baselines.

Between different kinds of data augmentation, CutMix in the specified configuration demonstrates similar clean accuracy as compared to baselines on CIFAR-10 yet significantly worse mean corruption accuracy on CIFAR-10-C. In other words improvements in in-distribution performance do not imply robustness to distribution shift. Pseudo-labeling with GAN samples brings moderate improvements to mean corruption accuracy in comparison with the baseline and Mixup model while maintaining close-to-baseline values of clean accuracy. Overall, the findings demonstrate that data augmentation designed to improve robustness cannot be treated as an off-the-shelf module.

Instead, the choice and strength of augmentation should be jointly considered with other hyperparameters including the optimizer and the number of epochs and be evaluated based on a realistic benchmark like CIFAR-10-C rather than clean test accuracy only. For further research it would make sense to repeat the analysis with another architecture (such as Wide ResNet), apply it to other datasets, consider more sophisticated generative models while developing automated techniques of tuning data augmentation and other training hyperparameters for improved robustness.

References

[1] A. Rebuffi, S. Gowal, D. K. Duvenaud, P. Kohli, and O. Vinyals, “Data augmentation can improve robustness,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2021.

[2] E. Mintun, C. Zhang, S. De, and S. Sra, “On interaction between augmentations and corruptions in natural corruption robustness,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2021.

[3] Z. Zhao, Z. Zhang, T. Chen, S. Khanna, and G. Hinton, “Image augmentations for GAN training,” arXiv:2006.02595, 2020.

[4] T. Karras, M. Aittala, J. Hellsten, S. Laine, J. Lehtinen, and T. Aila, “Training generative adversarial networks with limited data,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020.

[5] S. K. Y. Hong, T. Salimans, H. Zhang, and D. P. Kingma, “Differentiable augmentation for data-efficient GAN training,” in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2020.

[6] A. F. M. Agustsson, F. Mentzer, M. Tschannen, and E. Agustsson, “Masked generative adversarial networks are data-efficient learners,” arXiv, 2020.

[7] S. Khan, A. Sohail, and A. S. Qureshi, “A comprehensive review on deep convolutional neural networks,” IEEE Trans. Neural Netw. Learn. Syst., 2022.

[8] J. Yang, X. Chen, and L. Zhang, “GAN-based data augmentation for image classification: A survey and empirical study,” Eng. Appl. Artif. Intell., 2023.

[9] Y. Zhang, H. Wang, and J. Li, “A survey of mix-based data augmentation: Taxonomy, methods, applications, and explainability,” ACM Comput. Surv., 2024.

[10] S. El-Fakharany, A. M. Sarhan, and A. M. El-Gayar, “Image classification using deep learning: A comparative study,” in Proc. IEEE Int. Conf., 2021.